Knowledge Base Article

KB-017 / Rev 1.2

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Purpose and assessment context

A structured instructor-authored model for CIT-205. It illustrates how technical work becomes verifiable evidence. All incidents, measurements, approvals and names below are fabricated for teaching.

Article metadata

Title: Diagnose DNS name-resolution failures. Category: Networking / Client Connectivity. Audience: entry-level service desk technicians. Owner: CIT-205 learning KB.

Symptoms

Device can communicate with an approved IP endpoint but cannot open a training website by hostname. Record exact error, time, and whether issue affects one or many users.

Safety & prerequisites

Authorized workstation and terminal. Do not collect passwords. Do not alter production DNS or firewall settings without change approval.

Procedure 1

Record affected application and scope. Check local link status and assigned address; distinguish wired/Wi-Fi state.

Procedure 2

Test gateway connectivity. Test approved IP endpoint. Perform a DNS lookup of an approved training hostname; capture sanitized outputs.

Procedure 3

Compare observed resolver configuration against published approved baseline. If unauthorized or broad outage, stop and escalate.

Procedure 4

Only with approval, restore authorized configuration, keep rollback record, repeat all tests, and obtain user confirmation.

Expected results

Working IP path plus failed hostname lookup suggests DNS investigation; simultaneous failure suggests wider connectivity issue; one-host failure suggests host-specific issue.

Escalation & rollback

Escalate multiple impacted devices, inconsistent authoritative DNS, unauthorized settings, or unsuccessful remediation. Restore original approved values on rollback.

Evidence and verification matrix

STEP	EXPECTED SIGNAL	IF NOT OBSERVED	EVIDENCE
Gateway reachability	Replies	Inspect local path	Ping log
IP connectivity	Response	Check route/service	Test log
DNS lookup	Name resolves	Compare resolver	Lookup output
Application test	Page loads	Escalate service issue	User confirmation

Reviewer handoff and quality assurance

Document: KB-017 / Rev 1.2 | Revision: 1.2 | Classification: public instructional example

Acceptance checklist

- ☐ Relevant user / system impact is defined without publishing confidential information.
- ☐ Expected outcomes are distinguished from the technician's actual observations.
- ☐ Every conclusion is supported by a specific test or appropriate evidence reference.
- ☐ Changes are authorized, reversible when appropriate, and documented.
- ☐ Closure includes verification, remaining risk, escalation, and reviewer disposition.
- ☐ Accessible language is used; screenshots and supporting files are safe for public release.

Evidence register (model references)

KB-017 / Rev 1.2 E-01 | Initial report or baseline. Synthetic exhibit reference, supplied to model documentation. Students replace with genuine, redacted records.

KB-017 / Rev 1.2 E-02 | Troubleshooting observations. Synthetic exhibit reference, supplied to model documentation. Students replace with genuine, redacted records.

KB-017 / Rev 1.2 E-03 | Change / review / authorization notes. Synthetic exhibit reference, supplied to model documentation. Students replace with genuine, redacted records.

KB-017 / Rev 1.2 E-04 | Final validation and user confirmation. Synthetic exhibit reference, supplied to model documentation. Students replace with genuine, redacted records.

Instructor review

What works: the record captures context, action, evidence and outcome. Before qualifying real student work, reviewers must inspect supporting artifacts and verify that any technical claim or metric is true.

Fictional review disposition: EXEMPLAR APPROVED FOR TEACHING ONLY.

Reflection and next action

The value of this artifact is not the final checkbox; it is the trace from an observed symptom through a defensible decision to validated results. A real student should explain one revision made after feedback and link the relevant commit.